

JIDAPA CHANJAROEN

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RESEARCH PROFILE

Researcher with a background in industrial engineering and supply chain analysis, and research experience in biomass energy planning, agricultural residue supply chains, and techno-economic assessment of energy-intensive industries. My work focuses on integrating quantitative analysis, forecasting, and system-level supply chain assessment to support infrastructure and energy planning. I am seeking to further develop this research through doctoral study in energy systems and sustainable supply chains.

RESEARCH INTERESTS AND FUTURE DIRECTIONS

- Biomass energy and renewable energy systems
- Techno-economic analysis of energy-intensive industries
- Supply chain system analysis with a focus on demand forecasting
- Regional and spatial energy and infrastructure planning

EDUCATION

Master of Engineering (Industrial Engineering)
Chiang Mai University, Thailand

Nov 2020 – Jul 2024

Thesis: Machine Learning Application to Supply Chain Demand Forecasting of Agro-Residues for Biomass Power Plants

- Developed a quantitative framework integrating agricultural forecasting and supply chain analysis to assess regional biomass resource readiness.
- Applied time-series methods and machine learning as supporting tools to reduce forecasting uncertainty.
- Conducted a spatial assessment of agricultural residue availability and biomass heating value potential across 17 provinces in Northern Thailand.

Bachelor of Engineering (Industrial Engineering)
Chiang Mai University, Thailand

Aug 2016 – Jun 2020

Senior Project: Design of a contractor logistics performance measurement system for the coal mining industry

- Developed logistics performance indicators and evaluation metrics for contractor operations.
- Applied system analysis and operational performance assessment to support decision-making in logistics.

RESEARCH EXPERIENCE

Research Unit Member

2020 – Present

Supply Chain and Engineering Management Research Unit (SCEM)

Department of Industrial Engineering, Faculty of Engineering, Chiang Mai University

- Joined the Supply Chain and Engineering Management Research Unit (SCEM) as a research intern in 2020 and continued as a research assistant and research unit member throughout the master's program.
- Contributed to academic research and consultancy projects led by faculty supervisors, with responsibilities including data analysis, system-level assessment, and research reporting.
- Assisted in the preparation of analytical and technical reports for industry-oriented research projects.
- Supported academic research, consultancy projects, and teaching-related activities under the SCEM Research Unit and the Department of Industrial Engineering.

Research Assistant

Jul 2025 – Present

Prospect Supply Chain for Small Battery Electric Vehicles (BEV) in the Lancang–Mekong Region

- Conducting supply chain and system analysis for small battery electric vehicle (BEV) deployment in the Lancang–Mekong region.
- Supporting data analysis and analytical reporting for regional and cross-border electric mobility research.

Project Engineer / Researcher

Jun 2024 – Jun 2025

Strategic Assessment of Industrial Cluster Development under the BCG Economy Model

- Supported policy-oriented research on industrial cluster development under Thailand's BCG (Bio–Circular–Green) Economy framework, with involvement in project coordination, inter-regional data integration, and research progress monitoring across four regions.
- Assisted faculty supervisors by preparing analytical summaries and background information to support system-level assessment, stakeholder discussions, and planning-oriented research outputs.

PUBLICATIONS & MANUSCRIPTS

Journal Article (Published)

Chanjaroen, J., Jangkrajarn, V., Chernbumroong, S., Tippayawong, K.Y., Nakkiew, W., Boonprasope, A., & Premphet, P. (2026). Techno-economic feasibility of rail-steel manufacturing under fragmented infrastructure demand. *Energy Strategy Reviews*, 65, 102254. <https://doi.org/10.1016/j.esr.2026.102254>

METHODS & TOOLS

- Supply chain analysis and system-level assessment
- Techno-economic analysis (TEA)
- Time-series forecasting
- Machine learning (supporting forecasting analysis)
- Microsoft Excel and Python (for research analysis)

ACADEMIC ACTIVITIES

- **Operations Research Network Conference (OR-NET 2024)** – Oral Presentation, 2024 (Peer-reviewed national conference; paper published in TCI Tier 1 journal)
- **The 29th Tri-U International Joint Seminar and Symposium (Tri-U 2023)** – Participant, 2023
- **Summer Camp 2023**, College of Engineering, National Chung Hsing University, Taiwan – Participant, 2023
- **International Engineering Design Challenge (IEDC 2023)** – Participant, 2023

LANGUAGES

- Thai: Native
- English: Intermediate